

Module 1: Physical Layer & Signals

Network Topologies, OSI vs TCP/IP, Nyquist & Shannon Capacity, Line Coding & FDM/TDM

Module I: Data Communication & Networking Overview — Complete 13-Topic Tracker

Topic 1: Communication Model (Sender, Message, Medium, Protocol)

Topic 3: Networks (PAN, LAN, MAN, WAN Taxonomies)

Topic 5: OSI 7-Layer Reference Model & Layer Functions

Topic 7: Standards & Protocol Layers (Modularity & Encapsulation)

Topic 9: Data Transmission Concepts (Bit vs. Baud, Latency = $T_t + T_p + T_q + T_{pr}$)

Topic 11: Digital Data Transmission (Discrete Levels vs. Analog)

Topic 13: Channel Capacity (Nyquist Noiseless vs. Shannon Noisy Capacity)

Topic 2: Data Communications (Simplex, Half-Duplex, Full-Duplex)

Topic 4: The Internet (Packet Switching & TCP/IP Suite)

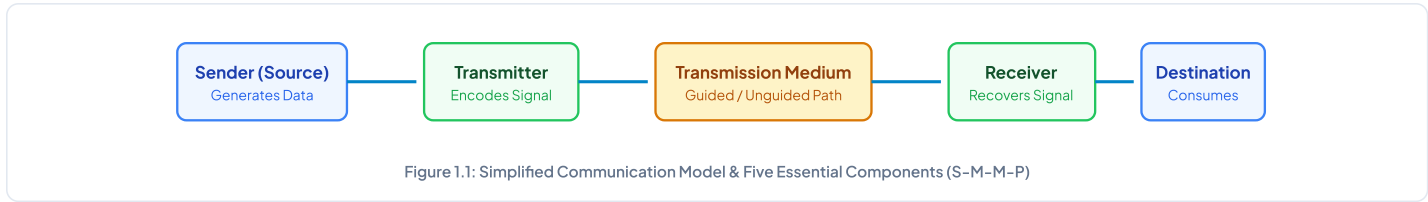
Topic 6: TCP/IP 4-Layer Protocol Architecture (OSI vs. TCP/IP)

Topic 8: Internet Applications (HTTP, SMTP, DNS, FTP, DHCP)

Topic 10: Analog Data Transmission (Amplitude, Frequency, Phase)

Topic 12: Transmission Impairments (Attenuation, Distortion, Noise)

Topic 1 & 2: Communication Model & Transmission Modes



Memory Hook: S-M-M-P
Sender → Message → Medium → Protocol (Syntax, Semantics, Timing)

Data Transmission Modes [UPLOADED PYQ]:

Mode	Directional Flow & Channel Mechanics	Real-World Example
1. Simplex	Unidirectional communication ($A \rightarrow B$). Only one device transmits; the other only receives. Channel capacity is 100% dedicated to one direction.	Keyboard to CPU, Television/Radio broadcast.
2. Half-Duplex	Bidirectional communication ($A \rightarrow B$ or $A \leftarrow B$), but not simultaneously . Both can transmit, but must alternate turns.	Walkie-Talkie systems.
3. Full-Duplex	Simultaneous bidirectional transmission ($A \rightleftarrows B$). Both stations can transmit and receive concurrently by dividing bandwidth.	Telephone networks, modern switched Ethernet.

Topic 3 & 4: Network Types & The Global Internet

Network	Geographic Coverage	Ownership & Speed	Representative Example
PAN	≈ 10 meters	Private; moderate speed (Bluetooth / Zigbee).	Phone ↔ Smartwatch ↔ Earbuds
LAN	Single room, office, or campus (≤ 1 km).	Privately owned; high data rate (1 Gbps–10 Gbps).	University campus / Lab LAN
MAN	City or metropolitan area (5–50 km).	Public/Private consortium; high speed.	Cable TV network across a city
WAN	Country, continent, or global (> 100 km).	Multiple telecom operators; variable speed.	The Global Internet

Topic 5 & 6: OSI 7-Layer Model vs. TCP/IP Architecture [UPLOADED PYQ]

OSI Layer	TCP/IP Layer	PDU Name	Core Functional Responsibility
7. Application	1. Application Layer	Data / Message	Provides network services to user apps (HTTP, SMTP, DNS, FTP).
6. Presentation			Syntax translation, data encryption/decryption (TLS), data compression.
5. Session			Dialog control, session checkpointing, synchronization tokens.
4. Transport	2. Transport Layer	Segment (TCP) / Datagram (UDP)	Process-to-process delivery, port addressing, flow control, reliability.
3. Network	3. Internet Layer	Packet / Datagram	Host-to-host routing, logical IP addressing, packet forwarding (IP, ICMP).
2. Data Link	4. Network Access Layer	Frame	Hop-to-hop framing, physical MAC addressing, CRC error detection, MAC access.
1. Physical		Raw Bits ('0'/'1')	Raw bit transmission over physical media, voltage levels, connectors.

🧠 Memory Trick for OSI 7 Layers

All People Seem To Need Data Processing

(Application → Presentation → Session → Transport → Network → Data Link → Physical)

Topic 7 & 8: Standards, Encapsulation & Internet Applications

Encapsulation occurs when data travels down the protocol stack at the sender: each layer prepends a protocol header containing control metadata:

Data $\xrightarrow{\text{Transport}}$ TCP Header + Data (Segment) $\xrightarrow{\text{Network}}$ IP Header + Segment (Packet) $\xrightarrow{\text{Data Link}}$ Frame Header + Packet + CRC Trailer (Frame)

Topic 9 – 11: Data Transmission Concepts, Latency & Digital Signals

1. Total End-to-End Latency Formula:

$$\text{Latency} = T_{\text{trans}} + T_{\text{prop}} + T_{\text{proc}} + T_{\text{queue}}$$

- **Transmission Delay** (T_{trans}): Time to push packet bits onto physical link: $T_{\text{trans}} = \frac{L \text{ (bits)}}{B \text{ (bps)}}$. - **Propagation Delay** (T_{prop}): Time for a bit to physically travel across distance: $T_{\text{prop}} = \frac{d \text{ (meters)}}{v \text{ (speed of light in medium)}}$. - **Bit Rate vs. Baud Rate**: Bit Rate = Baud Rate $\times \log_2(M)$ bps, where M is number of distinct signal voltage levels.

Topic 12: Transmission Impairments [UPLOADED PYQ]

Impairment	Physical Mechanism & Impact	Engineering Mitigation
1. Attenuation	Signal energy loss over distance due to medium resistance. Signal power drops logarithmically (dB).	Periodic Amplifiers (analog) or Regenerative Repeaters (digital).
2. Distortion	Signal changes shape because different frequency harmonic components travel at different phase velocities.	Equalizers and phase delay compensators.
3. Noise	Unwanted extraneous electrical signals injected into the channel: • <i>Thermal Noise</i> : Agitation of electrons ($N = kTB$). • <i>Crosstalk</i> : Electromagnetic coupling between adjacent wire pairs. • <i>Impulse Noise</i> : Spikes from lightning, power lines.	Shielding (STP/Coax), differential twisted pairs, optical fiber.

Topic 13: Channel Capacity Theorems [UPLOADED PYQ]

1. Nyquist Bit Rate (For Noiseless Channels):

$$C = 2 \times B \times \log_2(L) \text{ bps}$$

Where B is channel bandwidth in Hertz (Hz) and L is number of discrete signal voltage levels.

2. Shannon Channel Capacity (For Noisy Channels with Thermal Noise):

$$C = B \times \log_2(1 + \text{SNR}) \text{ bps}$$

Where $\text{SNR} = 10^{\frac{\text{SNR}_{\text{dB}}}{10}}$ is the linear Signal-to-Noise Ratio.

 [UPLOADED PYQ] Numerical: Shannon Capacity for Bandwidth 4 kHz and $\text{SNR}_{\text{dB}} = 24 \text{ dB}$

1. Linear $\text{SNR} = 10^{\frac{24}{10}} = 10^{2.4} \approx \mathbf{251.19}$.
2. $C = 4000 \times \log_2(1 + 251.19) = 4000 \times \log_2(252.19) \approx 4000 \times 7.978 = \mathbf{31,913 \text{ bps}}$ ($\approx 32 \text{ kbps}$).

 M1 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Compare OSI and TCP/IP models across 5 engineering parameters. (8 Marks)

1. **Layer Count:** OSI has 7 layers; TCP/IP has 4 (or 5) layers.
2. **Origins:** OSI was designed by ISO as a formal conceptual reference model before protocols were written; TCP/IP was designed pragmatically around actual working protocols (ARPANET).
3. **Services vs Protocols:** OSI strictly separates Services, Interfaces, and Protocols; TCP/IP combines services into protocols directly.
4. **Network Layer Communication:** OSI supports both Connectionless and Connection-Oriented at Network layer; TCP/IP supports only Connectionless (IP).
5. **Transport Layer Communication:** OSI supports only Connection-Oriented; TCP/IP supports both Connection-Oriented (TCP) and Connectionless (UDP).